

LUCKNOW, India

WMO: 423690

Lat: 26.761N

Lon: 80.889E

Elev: 125

StdP: 99.83

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	7.9	2.2	4.5	21.7	3.9	5.1	19.6	6.4	19.8	5.6	19.9	0.6	270	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.4	42.8	23.4	41.2	23.5	39.9	23.7	29.8	34.4	29.3	33.8	28.9	33.3	3.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.0	26.1	32.0	28.2	24.8	31.4	28.0	24.5	31.2	100.7	34.4	97.9	33.4	95.5	33.5	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.6	4.8	DB	4.1	44.7	1.4	1.5	3.0	45.7	2.2	46.6	1.4	47.5	0.4	48.5	
			WB	3.7	31.0	1.2	1.3	2.8	31.9	2.2	32.6	1.5	33.3	0.7	34.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	14.7	18.9	24.2	30.1	32.8	32.9	30.2	29.9	29.3	26.6	21.6	16.6
	DBStd	6.51	2.73	2.60	2.95	2.54	2.31	2.79	1.85	1.47	1.67	2.22	2.04	2.56
	HDD10.0	3	3	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	200	114	21	1	0	0	0	0	0	0	0	1	63
	CDD10.0	5729	149	249	441	604	707	687	628	616	578	516	348	205
	CDD18.3	2885	3	37	183	354	449	437	369	358	328	258	99	9
	CDH23.3	38391	38	386	2124	5142	7036	6745	4872	4568	3943	2616	818	104
Wind	WSAvg	21384	2	95	1078	3279	4692	4395	2459	2153	1781	1209	237	4
	WSAvg	1.8	1.5	1.8	2.0	2.3	2.4	2.5	2.2	2.0	1.8	1.0	0.9	1.1
	PrecAvg	1038	16	12	8	6	21	106	304	317	191	42	5	10
	PrecMax	2139	66	56	42	36	72	336	762	654	570	288	42	80
	PrecMin	493	0	0	0	0	0	0	106	45	15	0	0	0
Precipitation	PrecStd	420	18	16	11	9	21	91	175	150	144	77	11	19
	DB	26.2	32.0	38.2	43.0	44.9	44.9	39.1	37.1	37.0	36.2	32.0	27.1	
	MCWB	17.1	21.0	20.8	21.0	23.8	25.1	28.3	29.5	28.0	25.8	20.7	17.5	
	DB	24.2	29.1	36.0	41.2	42.9	42.8	36.9	35.2	35.2	34.9	30.2	25.9	
	MCWB	15.8	19.0	20.9	21.1	23.3	24.8	28.4	28.8	27.5	24.8	20.1	16.7	
Mean Coincident Wet Bulb Temperatures	DB	22.8	27.3	34.1	39.9	41.3	41.0	35.2	34.1	34.2	33.2	29.2	24.5	
	MCWB	14.9	17.8	20.3	21.3	23.4	25.2	28.3	28.5	27.3	24.2	19.6	16.3	
	DB	21.0	25.8	32.2	38.2	40.0	39.1	34.0	33.1	33.2	32.1	28.0	23.0	
	MCWB	14.4	17.0	19.6	21.3	23.5	25.9	28.3	28.2	27.0	23.6	18.9	15.6	
	WB	18.7	21.7	23.5	26.7	29.4	30.2	30.5	30.8	29.7	28.5	23.1	19.7	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	24.3	28.4	34.1	36.4	37.8	37.4	34.8	34.5	34.0	33.3	28.3	24.5	
	WB	17.2	20.1	22.2	25.2	28.3	29.2	29.7	29.8	29.0	27.1	21.9	18.3	
	MCDB	22.0	27.3	32.8	35.2	37.0	36.1	33.7	33.4	33.2	31.5	27.3	23.2	
	WB	16.1	19.0	21.2	24.2	27.2	28.7	29.2	29.2	28.4	26.2	21.0	17.3	
	MCDB	20.4	25.6	31.4	34.1	35.6	35.1	33.1	32.8	32.1	30.4	26.9	22.2	
Mean Daily Temperature Range	WB	15.2	17.9	20.4	23.2	26.3	28.1	28.7	28.7	27.9	25.5	20.2	16.4	
	MCDB	19.3	23.7	30.1	33.8	35.1	34.0	32.4	32.0	31.3	29.6	25.8	21.3	
	MDBR	11.6	13.3	15.2	15.7	13.4	10.1	6.6	6.4	7.4	11.3	13.7	12.6	
	5% DB	14.5	15.0	16.7	17.4	15.6	13.3	8.5	7.9	9.1	11.9	14.1	14.3	
	MCWBR	6.9	6.2	5.7	5.7	5.2	4.2	3.4	3.4	3.4	4.4	5.6	6.6	
Clear-Sky Solar Irradiance	5% WB	12.0	13.7	15.6	14.3	12.5	9.8	7.3	7.2	7.7	9.4	12.4	12.5	
	MCWBR	6.5	6.4	5.9	5.8	5.3	4.0	3.6	3.6	3.5	4.1	5.5	6.6	
	taub	0.565	0.533	0.520	0.598	0.695	0.744	0.674	0.615	0.590	0.662	0.674	0.638	
	taud	1.705	1.828	1.894	1.758	1.610	1.561	1.710	1.861	1.886	1.653	1.571	1.593	
	Ebn at Noon	658	737	781	733	664	629	673	711	714	621	563	570	
All-Sky Solar Radiation	Edh at Noon	208	198	195	229	266	277	239	204	194	233	237	225	
	RadAvg	3.01	4.41	5.82	6.60	6.68	5.72	4.57	4.69	4.76	4.61	3.69	2.99	
	RadStd	0.45	0.36	0.20	0.24	0.32	0.58	0.39	0.37	0.42	0.26	0.22	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page